

Lab: Bio-Inspired Sensor Design and Predictive Perception

Objective

Design bio-inspired sensing systems for robots by studying biological sensory organs and neural processing. You will compare biological and artificial perception strategies, implement a predictive coding-inspired sensor processing pipeline, and analyze how biological sensor design principles improve robotic perception.

Prerequisites

- Completion of the corresponding module.md lecture material
- Familiarity with Active Inference concepts (generative models, free energy, prediction errors)
- Basic pseudocode and diagram skills

Part 1: Biological Sensor Analysis

Compare biological and robotic sensing for three modalities:

1. **Vision:** Compare the vertebrate retina (on-center/off-surround, saccades, foveation) with a standard RGB camera. How does retinal processing implement prediction error computation at the sensor level?
2. **Touch:** Compare human fingertip mechanoreceptors (Merkel cells, Meissner corpuscles, Pacinian corpuscles) with robotic tactile sensors. What information is lost in current robotic touch?
3. **Proprioception:** Compare muscle spindles and Golgi tendon organs with joint encoders and current sensors. How does biological proprioception implement Active Inference?

Write your response here

Part 2: Predictive Coding Pipeline

Design a perception pipeline inspired by the brain's predictive coding architecture:

1. Define a hierarchical processing stack with at least three levels (e.g., edge detection, object boundaries, scene understanding).
2. Specify top-down predictions flowing from higher to lower levels.
3. Specify bottom-up prediction errors flowing from lower to higher levels.
4. How does this architecture differ from standard feedforward image processing?

Write your response here

Part 3: Active Sensing

Biological organisms actively control their sensors. Design active sensing strategies:

1. **Saccadic vision:** Design a camera control strategy inspired by eye saccades that fixates on informative regions.
2. **Whisking:** Design a tactile exploration strategy inspired by rodent whisking for shape recognition.
3. **Echolocation:** Design an ultrasonic sensing strategy inspired by bat echolocation for obstacle mapping.

For each, how does Active Inference (expected free energy minimization) explain the biological sensing strategy?

Write your response here

Part 4: Bio-Inspired Sensor Design Table

Propose bio-inspired improvements to standard robotic sensors:

Standard Sensor	Biological Inspiration	Proposed Improvement	AI Benefit
RGB camera	Foveal vision		
Pressure sensor	Fingertip mechanoreceptors		
Microphone	Barn owl auditory system		
LIDAR	Bat echolocation		

Write your response here

Summary Table

Concept	Classical Robotics	Active Inference	Your Design
Core mechanism			
Uncertainty handling			
Adaptation			
Key advantage			

References

- Lanillos, P., et al. (2021). Active Inference in Robotics and Artificial Agents. *Frontiers in Neurorobotics*.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press.
- Pio-Lopez, L., Nizard, A., Friston, K., & Pezzulo, G. (2016). Active Inference and robot control. *Journal of the Royal Society Interface*, 13(122).